

Exact Finite-Level Mixing Geometry in Collatz Dynamics

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Abstract

We study the distribution of accelerated Collatz orbits modulo powers of two. The Syracuse map

$$T(n) = \frac{3n+1}{2^{v_2(3n+1)}}$$

induces a finite Markov kernel P_K on the odd residue classes modulo 2^K by lifting a residue class to the odd 2-adic integers with Haar measure and reducing the image modulo 2^K . We prove that the support of $P_K^t(a, \cdot)$ has an exact dyadic arithmetic-progression geometry at every pre-mixing step. This yields a closed formula for the total variation distance from the uniform distribution and the exact support-mixing time

$$T_{\text{mix}}(K) = K - 1,$$

achieved by the worst-case class $a = 2^K - 1$.

We also describe the adjoint operator $\mathcal{L}_K = P_K^*$ as a normalized finite inverse-tree transfer operator. The finite-level dual representation contains $K - 1$ ordinary inverse branches together with one tail branch collecting all valuation levels $v \geq K$. This normalization is essential: the finite operator is stochastic, while the infinite 2-adic limiting expression has the geometric weights 2^{-v} .

The paper does not claim a proof of the Collatz conjecture. Its main contribution is an exact finite-level geometry and a precise formulation of the remaining population-to-orbit obstruction. Numerical experiments are consistent with a power-law spectral-gap scaling, but the exponent and its interpretation remain open.

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1 Introduction

The Collatz conjecture asserts that every positive integer eventually reaches 1 under iteration of the map $n \mapsto n/2$ for even n and $n \mapsto 3n + 1$ for odd n . A standard way to remove the deterministic divisions by two is to study the Syracuse map T , which maps odd integers to odd integers.

This paper studies the finite quotients of this map modulo 2^K . The guiding question is not whether this finite analysis proves the Collatz conjecture. It does not. The point is to isolate, exactly, what finite 2-adic mixing does prove and where the orbitwise obstruction remains.

Important clarification. This paper proves exact finite-level statements modulo powers of two. It does not prove the Collatz conjecture, orbitwise ergodicity, asymptotic spectral rigidity, or convergence of all trajectories.

Main results

Let Ω_K be the set of odd residue classes modulo 2^K , and let π_K be the uniform probability measure on Ω_K . The paper establishes:

1. exact dyadic arithmetic-progression support geometry for $P_K^t(a, \cdot)$;

2. an exact total variation formula;
3. the exact support-mixing time $T_{\text{mix}}(K) = K - 1$;
4. a normalized finite dual representation of P_K^* as an inverse-tree transfer operator;
5. a precise open problem concerning spectral rigidity and orbitwise lifting.

The main theorem can be summarized as follows.

Theorem 1.1 (Exact finite-level geometry, informal). *For every $K \geq 2$, every $a \in \Omega_K$, and every $t \geq 0$, the law $P_K^t(a, \cdot)$ is uniform on a dyadic arithmetic progression inside Ω_K . Its size doubles according to the accumulated 2-adic valuation until it fills all of Ω_K . Consequently the total variation distance from π_K is known exactly, and mixing occurs precisely when the progression fills all odd residue classes.*

Relation to prior work

Probabilistic and stopping-time approaches to the $3x + 1$ problem go back at least to Terras and are surveyed by Lagarias. The present paper is concerned with exact finite quotient geometry rather than asymptotic density estimates. It complements earlier structural work by giving an explicit finite Markov-kernel description and a closed total-variation formula.

2 Setup and notation

2.1 The finite state space

Fix $K \geq 2$ and set

$$\Omega_K = \{a \in \mathbb{Z}/2^K\mathbb{Z} : a \text{ is odd}\}, \quad |\Omega_K| = 2^{K-1}.$$

Let π_K denote the uniform probability measure on Ω_K .

2.2 The Markov kernel

Given an odd residue class $a \in \Omega_K$, choose an odd 2-adic integer $n \in \mathbb{Z}_2^\times$ with $n \equiv a \pmod{2^K}$ according to normalized Haar measure on that cylinder, apply the Syracuse map, and reduce modulo 2^K . This defines a Markov kernel P_K on Ω_K .

For an integer representative a of the class, put

$$r(a) = v_2(3a + 1), \quad h_K(a) = \min\{r(a), K - 1\}.$$

Then

$$P_K(a, \cdot) = \text{Uniform}\left(\text{AP}\left(B_K(a), 2^{K-h_K(a)}, 2^{h_K(a)}\right)\right),$$

where $B_K(a)$ is any odd element of the image support. If $r(a) < K$, one may take

$$B_K(a) = \frac{3a + 1}{2^{r(a)}} \pmod{2^K}.$$

If $r(a) \geq K$, the image support is already all of Ω_K , so the choice of the odd base $B_K(a)$ is immaterial.

Remark 2.1 (Why the truncation is necessary). The valuation tail $r(a) \geq K$ is a real finite-level issue. In that case the next image modulo 2^K is uniform on all odd residue classes. Thus the effective finite-level height is $K - 1$, not K or larger. This convention prevents the common normalization error in which the finite inverse-branch weights fail to sum to one.

2.3 Dyadic arithmetic progressions

For an odd base b , a dyadic spacing $d = 2^q$, and an integer m , write

$$\text{AP}(b, d, m) = \{b + jd \pmod{2^K} : j = 0, 1, \dots, m - 1\}.$$

All arithmetic progressions in this paper are considered modulo 2^K . The case $d = 2$ and $m = 2^{K-1}$ is the full odd state space Ω_K .

2.4 Valuation sums

Before the support has filled Ω_K , define a deterministic base orbit by

$$F_K(a) = \frac{3a + 1}{2^{r(a)}} \pmod{2^K} \quad (r(a) < K),$$

and set

$$V_s(a) = r(F_K^{s-1}(a)), \quad S_t(a) = \sum_{s=1}^t V_s(a).$$

The relevant finite-level support size is governed by the capped sum

$$H_t(a) = \min\{S_t(a), K - 1\}.$$

Once $H_t(a) = K - 1$, the distribution is already uniform on Ω_K , and subsequent choices of the deterministic base no longer matter.

3 Exact support geometry

3.1 A local valuation-stability lemma

Lemma 3.1 (Valuation stability on sufficiently fine residue classes). *Let c_0 be odd and let $v = v_2(3c_0 + 1)$. If*

$$c \equiv c_0 \pmod{2^{v+1}},$$

then

$$v_2(3c + 1) = v.$$

Proof. Write $c = c_0 + 2^{v+1}m$. Since $3c_0 + 1 = 2^v u$ with u odd,

$$3c + 1 = (3c_0 + 1) + 3 \cdot 2^{v+1}m = 2^v(u + 2 \cdot 3m).$$

The factor $u + 2 \cdot 3m$ is odd, so the 2-adic valuation is exactly v . □

Lemma 3.2 (API – arithmetic-progression invariance before mixing). *Suppose $P_K^t(a, \cdot)$ is uniform on*

$$\text{AP}(c_0, 2^{K-H}, 2^H)$$

with $H \leq K - 2$. Let $v = v_2(3c_0 + 1)$.

If $H + v \leq K - 1$, then every element of the support has the same next-step valuation v , and the next support is again a dyadic arithmetic progression of size 2^{H+v} .

If $H + v \geq K - 1$, then the next support is all of Ω_K .

Proof. If $H + v \leq K - 1$, then the common difference 2^{K-H} is divisible by 2^{v+1} . Hence every support point c satisfies $c \equiv c_0 \pmod{2^{v+1}}$, and Lemma 3.1 gives $v_2(3c + 1) = v$. The image of $c = c_0 + \ell 2^{K-H}$ is

$$\frac{3c + 1}{2^v} \equiv \frac{3c_0 + 1}{2^v} + 3\ell 2^{K-H-v} \pmod{2^K}.$$

Combining these translated progressions with the internal spacing 2^{K-v} gives a single dyadic progression whose spacing is

$$\gcd(2^{K-H-v}, 2^{K-v}) = 2^{K-H-v}$$

and whose size is 2^{H+v} .

If $H + v \geq K - 1$, the same gcd computation reaches spacing 2 or finer on the odd state space, so the support fills all of Ω_K . $\square$

3.2 Exact total variation formula

Theorem 3.3 (ETV – exact total variation formula). *For every $a \in \Omega_K$ and every $t \geq 0$, the distribution $P_K^t(a, \cdot)$ is uniform on a dyadic arithmetic progression of size $2^{H_t(a)}$ in Ω_K , where*

$$H_t(a) = \min\{S_t(a), K - 1\}.$$

Consequently

$$\text{TV}(P_K^t(a, \cdot), \pi_K) = 1 - 2^{H_t(a) - (K-1)}.$$

In particular, this quantity is zero exactly when $H_t(a) = K - 1$.

Proof. The statement follows by induction from Lemma 3.2. At $t = 0$, the support is the one-point progression $\text{AP}(a, 2^K, 1)$, so $H_0 = 0$. The induction step is precisely the two-case alternative in Lemma 3.2: either the support remains a dyadic progression with size multiplied by 2^v , or it reaches the full odd state space.

If the support has size 2^H , the law assigns mass 2^{-H} to each support point, while π_K assigns mass $2^{-(K-1)}$ to every odd residue class. For $H < K - 1$,

$$\begin{aligned} \text{TV}(P_K^t(a, \cdot), \pi_K) &= \frac{1}{2} [(2^{K-1} - 2^H)2^{-(K-1)} + 2^H(2^{-H} - 2^{-(K-1)})] \\ &= 1 - 2^{H-(K-1)}. \end{aligned}$$

For $H = K - 1$, the two distributions are equal. $\square$

4 Exact support-mixing time

Theorem 4.1 (Exact support-mixing time). *For the finite kernel P_K on Ω_K ,*

$$T_{\text{mix}}(K) = K - 1,$$

where mixing means equality with the uniform distribution in total variation.

Proof. Since every nonterminal valuation contributes at least 1, we have $H_{K-1}(a) = K - 1$ for every $a \in \Omega_K$. Thus every starting class reaches the uniform distribution in at most $K - 1$ steps.

For the lower bound, take $a = 2^K - 1$. Then

$$3a + 1 = 3 \cdot 2^K - 2 = 2(3 \cdot 2^{K-1} - 1),$$

so the first valuation is 1. The deterministic base remains congruent to -1 modulo decreasing powers of two, and the same computation gives valuation 1 for the first $K - 2$ steps. Hence $H_{K-2}(a) = K - 2 < K - 1$, so the distribution is not yet uniform at time $K - 2$. Therefore $T_{\text{mix}}(K) = K - 1$. $\square$

Remark 4.2 (Spectral consequences are not automatic). The exact total-variation mixing time by itself does not give the conjectural spectral-gap asymptotics. Standard finite-chain inequalities are consistent with the result above, but they do not prove a sharp lower bound of the form $\delta_K \geq CK^{-\alpha}$ with $\alpha < 1$. That sharper estimate is isolated below as an open spectral problem.

5 Numerical spectral observations

Let $\lambda_2(K)$ denote a numerically computed nontrivial eigenvalue of the finite transition matrix, and put $\delta_K = 1 - |\lambda_2(K)|$ for this experiment. Since the kernel is not presented here as a reversible self-adjoint chain, these eigenvalue computations should be interpreted as numerical evidence rather than as a theorem about ℓ^2 convergence.

5.1 Numerical data

Sparse-matrix computations are consistent with a power-law behavior

$$\delta_K \approx CK^{-\alpha}, \quad \alpha \approx 0.39,$$

over the tested range.

K	$ \Omega_K $	$ \lambda_2 $	δ_K	$\delta_K K^{0.39}$
8	128	0.255	0.745	1.19
10	512	0.282	0.718	1.23
12	2048	0.304	0.696	1.27
14	8192	0.352	0.648	1.25
16	32768	0.368	0.632	1.28
18	131072	0.417	0.583	1.24
20	524288	0.431	0.569	1.26

The apparent exponent is numerically stable in this range. Its theoretical meaning remains open.

5.2 Support mixing versus spectral decorrelation

The exact result $T_{\text{mix}}(K) = K - 1$ concerns support expansion and total variation. The numerical spectral data concern decay of eigenmodes or correlations. These are different notions. The central phenomenon is therefore

$$\text{support expansion} \neq \text{arithmetic decorrelation}.$$

This distinction is the main conceptual reason for separating the rigorous finite-level geometry from the open orbitwise problem.

6 Dual representation and inverse-tree structure

6.1 Finite inverse branches

For $1 \leq v \leq K - 1$, define the ordinary inverse branches

$$T_K^{-v}(b) = 3^{-1}(b2^v - 1) \pmod{2^K}, \quad b \in \Omega_K,$$

where 3^{-1} is the inverse of 3 modulo 2^K . Define the tail branch

$$T_K^{-\geq K}(b) = 3^{-1}(b2^K - 1) \pmod{2^K}.$$

The tail branch collects all lifted preimages with valuation at least K .

Lemma 6.1 (Finite inverse-branch identification). *For each $b \in \Omega_K$, the ordinary branches $T_K^{-v}(b)$ for $1 \leq v \leq K - 1$ have valuation level v and map to b . The tail branch $T_K^{-\geq K}(b)$ represents the unique residue class whose lifted preimages have valuation at least K and whose image may reduce to b modulo 2^K .*

Proof. For $1 \leq v \leq K - 1$, if $a = T_K^{-v}(b)$, then

$$3a + 1 \equiv b2^v \pmod{2^K}.$$

Since b is odd and $v < K$, this congruence has exact 2-adic valuation v modulo 2^K , and the Syracuse image is b modulo 2^K .

For the tail branch, the congruence is

$$3a + 1 \equiv b2^K \equiv 0 \pmod{2^K}.$$

Thus $3a + 1$ is divisible by 2^K . The Haar distribution on lifts of this residue class supplies the remaining valuation tail, whose total probability is $2^{-(K-1)}$ at the finite level. $\square$

6.2 The normalized dual operator

The Markov kernel acts on measures by right multiplication. Its adjoint on functions, with respect to counting measure, is denoted by $\mathcal{L}_K = P_K^*$.

Theorem 6.2 (Normalized finite dual representation). *For every function $f : \Omega_K \rightarrow \mathbb{R}$ and every $b \in \Omega_K$,*

$$(\mathcal{L}_K f)(b) = \sum_{v=1}^{K-1} 2^{-v} f(T_K^{-v}(b)) + 2^{-(K-1)} f(T_K^{-\geq K}(b)).$$

The weights sum to one.

Proof. The column of the transition matrix over a fixed b receives one contribution of weight 2^{-v} from each ordinary valuation level $1 \leq v \leq K-1$. The remaining Haar tail $v \geq K$ is represented by the single finite residue class $T_K^{-\geq K}(b)$ and has total weight

$$\sum_{v=K}^{\infty} 2^{-v} = 2^{-(K-1)}.$$

Therefore the total column weight is

$$\sum_{v=1}^{K-1} 2^{-v} + 2^{-(K-1)} = 1,$$

and the displayed formula follows. $\square$

Remark 6.3 (Infinite 2-adic limiting form). At the formal infinite 2-adic level, the tail is not truncated and one expects the operator

$$(\mathcal{L}_{\infty} f)(b) = \sum_{v=1}^{\infty} 2^{-v} f(3^{-1}(b2^v - 1)).$$

The finite theorem above is the normalized truncation of this expression, not the naive finite sum $\sum_{v=1}^K 2^{-v}$.

6.3 Spectral rigidity as an open problem

The eigenvalue equation for the finite adjoint operator is

$$\lambda f(b) = \sum_{v=1}^{K-1} 2^{-v} f(T_K^{-v}(b)) + 2^{-(K-1)} f(T_K^{-\geq K}(b)).$$

Thus a slowly decaying eigenmode must maintain coherence across many inverse 2-adic dilation scales simultaneously. This motivates the following open problem.

Open Problem 6.4 (Spectral rigidity). Prove or disprove that there exist constants $C > 0$ and $\alpha < 1$ such that, for an appropriate spectral gap or singular-value gap δ_K associated with P_K ,

$$\delta_K \geq CK^{-\alpha} \quad (K \geq 2).$$

7 Connection to the Collatz conjecture

7.1 What the finite geometry implies

Theorem 3.3 implies that if one samples a random odd 2-adic lift in a fixed residue class modulo 2^K , then after $K - 1$ applications of the Syracuse map its residue modulo 2^K is exactly uniform on Ω_K .

This is a rigorous population-level statement. It should not be confused with an orbitwise statement about a fixed positive integer.

7.2 The obstruction to lifting

The Collatz conjecture is an orbitwise assertion. The finite theorem says that populations of 2-adic lifts mix exactly at every fixed scale. A hypothetical divergent positive orbit would have to avoid being controlled by these finite-level population averages across all scales simultaneously.

More precisely, such an orbit would require persistent cross-scale arithmetic coherence across infinitely many 2-adic levels. The exact finite-level geometry identifies where that coherence would have to live, but it does not prove that such coherence is impossible.

8 Discussion

8.1 What is proved

The rigorous part of the paper is the exact finite-level support geometry:

- $P_K^t(a, \cdot)$ is uniform on a dyadic arithmetic progression until it fills Ω_K ;
- the total variation distance is exactly computable;
- the support-mixing time is exactly $K - 1$;
- the finite adjoint operator has a normalized inverse-tree representation.

8.2 What remains open

The following questions remain open:

- sharp spectral or singular-value estimates for P_K ;
- a theory of the limiting 2-adic transfer operator;
- orbitwise decorrelation along a fixed Collatz trajectory;
- any implication strong enough to prove or disprove the Collatz conjecture.

9 Summary of results

Result	Status	Location
Valuation stability lemma	Proved	Section 3
AP invariance before mixing	Proved	Section 3
Exact TV formula	Proved	Section 3
Exact support-mixing time $K - 1$	Proved	Section 4
Finite dual representation with tail branch	Proved	Section 6
Power-law spectral scaling	Numerical only	Section 5
Spectral rigidity	Open	Section 6
Orbitwise lifting	Open	Section 7

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